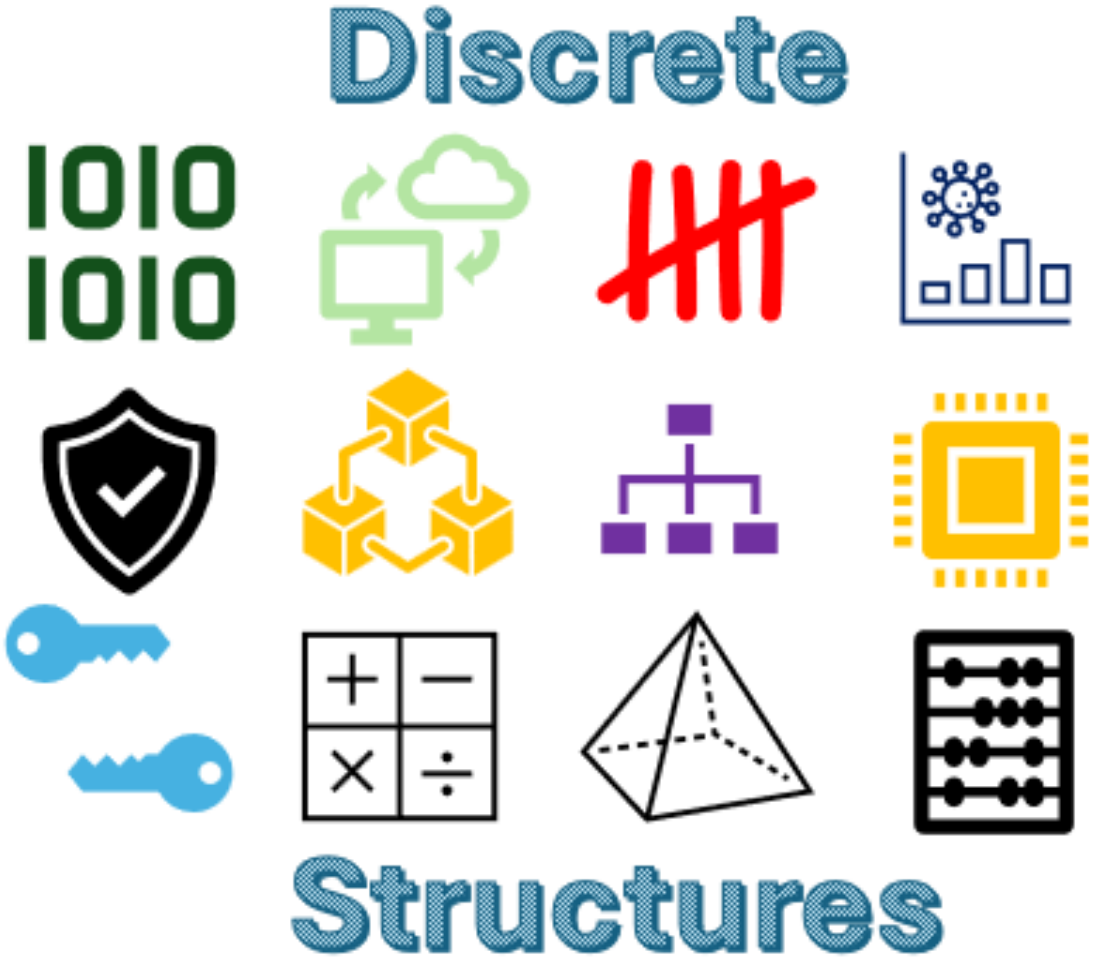
COMP 2310: Discrete Structures
Fall 2026

Chapter 2: Logic



Exact cover problem

- Combinatorial optimization problem
- Example:
  - Given a set A and a finite number of subsets of A, say $A_1, A_2, \ldots, A_k$, is it possible to find a collection S of the A_is with the following two properties:
    - Any two sets in S are disjoint (no common elements or $A \cap B = \emptyset$)
    - The union of all sets in S is A

Exact cover problem

- Let $A = \{a, b, c, d, e, f, g, h, i, j\}$ and $A_1 = \{a, c, d\}$, $A_2 = \{a, b, e, f\}$, $A_3 = \{b, f, g\}$, $A_4 = \{d, h, l\}$, $A_5 = \{a, h, j\}$, $A_6 = \{e, h\}$, $A_7 = \{c, i, j\}$, $A_8 = \{i, j\}$. Is there an exact cover of A with respect to $A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8$?

  $A_1 \cap A_3 \cap A_6 \cap A_8 = \emptyset$
  $A_1 \cup A_3 \cup A_6 \cup A_8 = A$

- Example scenario:
  - $A = \{a, b, c, d, e, f, g, h, i, j\}$ ➔ software development task including a→front-end design, b→back-end development etc.
  - Task is divided to different groups of developers ➔ Individual subsets
  - Apply the exact cover problem to validate whether all the tasks are assigned related to the development

Exact cover problem: using matrix

- Let $A = \{a, b, c, d, e, f, g, h, i, j\}$ and $A_1 = \{a, c, d\}$, $A_2 = \{a, b, e, f\}$, $A_3 = \{b, f, g\}$, $A_4 = \{d, h, l\}$, $A_5 = \{a, h, j\}$, $A_6 = \{e, h\}$, $A_7 = \{c, i, j\}$, $A_8 = \{i, j\}$. Is there an exact cover of A with respect to $A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8$?
- Matrix representations (rows→subsets, columns→elements)

$$\mathbf{M} = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}.$$

(a) $\mathbf{r}_i \wedge \mathbf{r}_j = \mathbf{0}$ for $1 \le i, j \le k$ and

(b) $\mathbf{r}_1 \vee \mathbf{r}_2 \vee \cdots \vee \mathbf{r}_k = \mathbf{1}$

Exact cover problem: using matrix

- Knuth's algorithm

Algorithm ALGORITHM X

Label the rows of the original matrix **A** and retain these labels throughout the application of the algorithm. Begin with $\mathbf{M} = \mathbf{A}$ and $L = \{\ \}$.

Step 1 If **M** has a column of 0's, then there is no solution, since by definition the rows representing an exact cover must together have 1's in every position.

Step 2 Otherwise:

(a) Choose the column **c** of **M** with the fewest 1's.

(b) Choose a row **r** with a 1 in column **c**, and place the number **r** in L.

(c) Eliminate any row $\mathbf{r}_1$ having the property that $\mathbf{r} \wedge \mathbf{r}_1 \neq \mathbf{0}$ (where **0** represents a row with all 0 entries).

(d) Eliminate all columns in which **r** has a 1.

(e) Eliminate row **r**.

(f) Let **M** be the resulting matrix and repeat steps 1 and 2. ●

- Example 4

Exercise

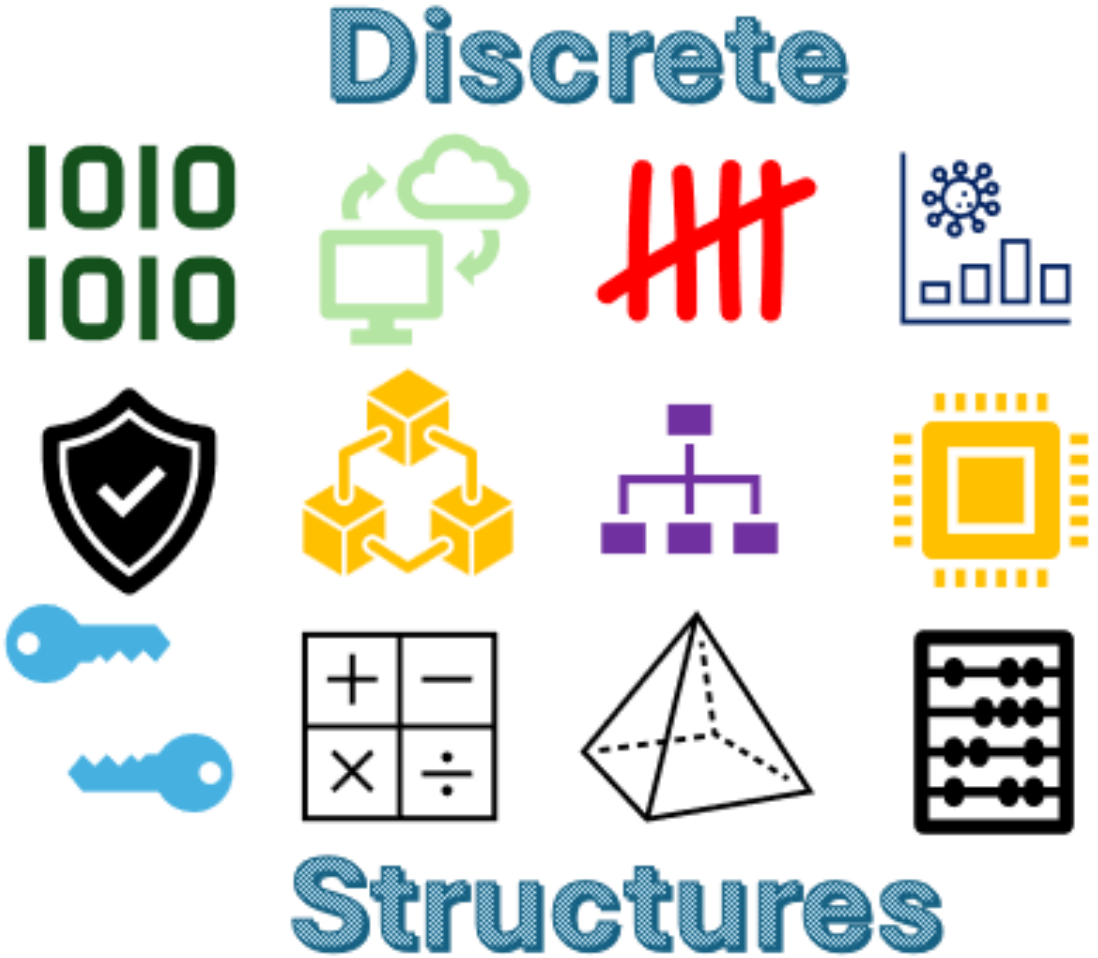
- **Company 1** = {Software Engineer, Data Scientist, Database Administrator}
- **Company 2** = {Machine Learning Engineer, Database Administrator, Game Developer, Web Developer}
- **Company 3** = {Software Engineer, Game Developer, Robotics Engineer}
- **Company 4** = {Cybersecurity Analyst, Systems Administrator, Robotics Engineer}
- **Company 5** = {Machine Learning Engineer, Systems Administrator}
- **Company 6** = {Software Engineer, Data Scientist, DevOps Engineer}

Use the exact cover algorithm to identify the companies to apply to in order to cover all unique job roles

Company 2, Company 4, Company 6

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Chapter 3: Counting

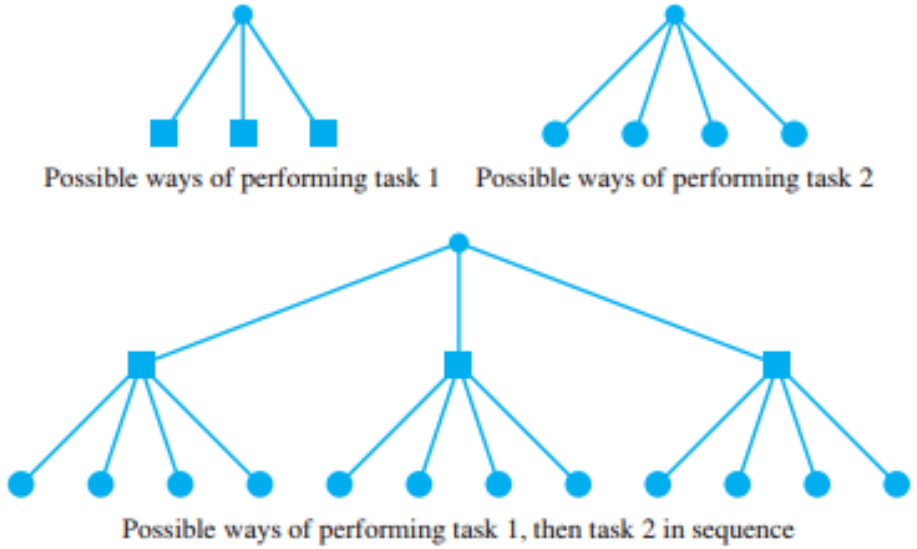


Looking back: history

- Please review

Permutation

- Permutation = arrangements
- Suppose that two tasks *T*1 and *T*2 are to be performed in sequence. If *T*1 can be performed in *n*1 ways, and for each of these ways *T*2 can be performed in *n*2 ways, then the sequence *T*1*T*2 can be performed in *n*1*n*2 ways.
- **Multiplication principle of counting**
- Example: Task 1 (n1 = 3) and Task 2 (n2 = 4)



Permutation

- **Extension:** Suppose that tasks $T_1, T_2, \ldots, T_k$ are to be performed in sequence. If T_1 can be performed in n_1 ways, and for each of these ways T_2 can be performed in n_2 ways, and for each of these $n1n2$ ways of performing T_1T_2 in sequence, T_3 can be performed in n_3 ways, and so on, then the sequence $T_1T_2 \cdot \cdot \cdot T_k$ can be performed in exactly $n_1n_2 \cdot \cdot \cdot n_k$ ways.

Permutation

- Example
  - A label identifier, for a computer system, consists of one letter followed by three digits. If repetitions are allowed, how many distinct label identifiers are possible?
    - **Format:** letter, digit, digit, digit
      - letter: A-Z = 26
      - Digit = 0 – 9 = 10
      - 26 * 10 * 10 * 10

Permutation

- How many different sequences, each of length *r*, can be formed using elements from *A* if
  - **(a)** elements in the sequence may be repeated?
  - **(b)** all elements in the sequence must be distinct?

Answer (a): assume r = 5, so the sequence format AAAAA. Assume, A can be any of the 26 alphabet. Therefore, for each position in the sequence, 26 possibilities possible. Therefore,

- 26*26*26*26*26
- 26^5
- A^r

Permutation

- How many different sequences, each of length *r*, can be formed using elements from *A* if
  - **(a)** elements in the sequence may be repeated?
  - **(b)** all elements in the sequence must be distinct?

Answer (b): assume r = 5, so the sequence format AAAAA.

Distinct sequences: n x (n-1) x (n-2) x (n-3) x ….. x (n-r + 1) = $\frac{n!}{(n-r)!}$

$26 \times 25 \times 24 \times 23 \times 22 = \frac{26!}{(26-5)!}$

Permutation

- How many three-letter “words” can be formed from letters in the set {*a, b, y, z*} if repeated letters are allowed?

$$4^3$$

Permutation

- If $1 \leq r \leq n$, then ${}_nP_r$ is the number of permutations of n objects taken r at a time, is $n \cdot (n - 1) \cdot (n - 2) \cdot \cdots \cdot (n - r + 1)$.
- Let A be {1, 2, 3, 4}. Then the sequences 124, 421, 341, and 243 are some permutations of A taken 3 at a time. The sequences 12, 43, 31, 24, and 21 are examples of different permutations of A taken two at a time. By Theorem 4, the total number of permutations of A taken three at a time is 4 P3 or 4·3·2 or 24. The total number of permutations of A taken two at a time is 4 P2 or 4 · 3 or 12.

Permutation

- How many “words” of three distinct letters can be formed from the letters of the word MAST?

- ${}_nP_r = \frac{n!}{(n-r)!}$

$$_4P_3 = 24$$

Permutation

- The number of distinguishable permutations that can be formed from a collection of *n* objects where the first object appears *k*1 times, the second object *k*2 times, and so on, is

$$\frac{n!}{k_1!\,k_2!\cdots k_t!}, \quad \text{where } k_1 + k_2 + \cdots + k_t = n.$$

- Distinguishable: ability to identify different arrangement → number of unique ways for arrangements

Permutation

- The number of distinguishable “words” that can be formed from the letters of **DISCRETE** is
  - D= 1
  - I = 1
  - S = 1
  - R = 1
  - C = 1
  - T = 1
  - E = 2

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$$\textit{Unique arrangements} = \frac{total\ arrangements}{overcount\ caused\ by\ identical\ objects}$$